Programming for Data Science

Course Introduction

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We do data science activities in daily life

Introduction

- You might not realize it, but you already use data science thinking in your daily life!
- Every time you *make an important decision*, you're following the same *workflow* that professional data scientists use.
- Examples:
  - **Choosing your university major/course**
  - Buying a new mobile phone
- Let's see how your natural decision-making process mirrors professional data science!

Choosing Your University Major

- **Step 1: Ask a Meaningful Question**
- Your Question: "Which major should I choose that fits my interests, skills, and career goals?“
- Why This Matters:
  - Defines **what** you're trying to solve
  - Sets clear **criteria for success**

- **In Data Science**: Without the right question, even perfect data is useless

Choosing Your University Major

- **Step 2: Collect Data**

  What Data Do You Gather?

- About Yourself:
  - Your favorite subjects in high school
  - Skills you enjoy using (math, writing, creativity, problem-solving)
  - Personality traits (prefer working alone or in teams?)
  - Long-term career aspirations

Choosing Your University Major

- **Step 2: Collect Data**

  What Data Do You Gather?

- About Majors:
  - Course descriptions and curriculum
  - Career opportunities and job market demand
  - Average salaries in different fields
  - Alumni success stories
  - University rankings for specific programs

Choosing Your University Major

- **Step 2: Collect Data**

  What Data Do You Gather?

- **Data Sources**:
  - University websites and brochures
  - Conversations with current students and professors
  - Career counseling sessions
  - Online forums and reviews
  - Job market reports
- **In Data Science**: Just like you gather information from multiple sources, data scientists collect data from databases, APIs, surveys, and external sources.

Choosing Your University Major

- **Step 4: Explore the Data**

Before cleaning anything, you look at what you've got:

- "I have salary data for 6 majors but not for the other 2."
- "Salary numbers from forum A are much higher than from official reports — suspicious?"
- "Most reviews for major X are from 5+ years ago."

- **In Data science**: exploration reveals what the data contains, and what's wrong with it.

Choosing Your University Major

- **Step 4: Preprocess (clean) the Data**
- Filter Out Irrelevant Information:
  - Ignore majors that don't interest you at all
  - Remove outdated salary information
  - Discard biased opinions from unreliable sources
- Organize Your Information:
  - Create a comparison table or spreadsheet
  - Group similar majors together
  - One row per major, one column per criterion
  - Note missing information (maybe you don't know job prospects for some fields)

Choosing Your University Major

- **Step 4: Preprocess (clean) the Data**
- Handle Inconsistencies:
  - Sources disagree on salary → take a reasonable average (or the median)
  - A friend's opinion conflicts with official data → weigh credibility
  - Missing information → look for additional sources, or note it as unknown
- **In Data Science**: Data scientists clean messy data, handle missing values, remove duplicates, and standardize formats before analysis.

Choosing Your University Major

- **Step 5: Analyze the Data**

Criterion (weight)	Computer Science	Business	Graphic Design
Interest (0.35)	8	7	9
Job market (0.25)	9	7	5
Salary (0.20)	8	7	5
Fit with my skills (0.20)	8	7	6
Weighted score	**8.25**	**7.00**	**6.60**

Choosing Your University Major

- **Step 5: Analyze the Data**
- Look for Patterns:
  - Notice that majors you're interested in share common themes
  - See correlations: "Majors with more math tend to have higher starting salaries“
  - Identify trends: "Tech-related fields are growing faster“

- **In Data Science**: Data scientists use statistical analysis, visualizations, and algorithms to find patterns, correlations, and insights in data.

Choosing Your University Major

- **Step 6: Decide & communicate**
- **Synthesise:** CS scores highest overall, matches your problem-solving skills, and has strong job growth.
  - **Decision:** *"I'll major in Computer Science with a minor in Business to combine my interests and keep my career options open."*
  - **Communicate:** explain it to your parents with your comparison table, not just "because I like it“.
- **In data science**: results are only useful if decision-makers understand and trust them.

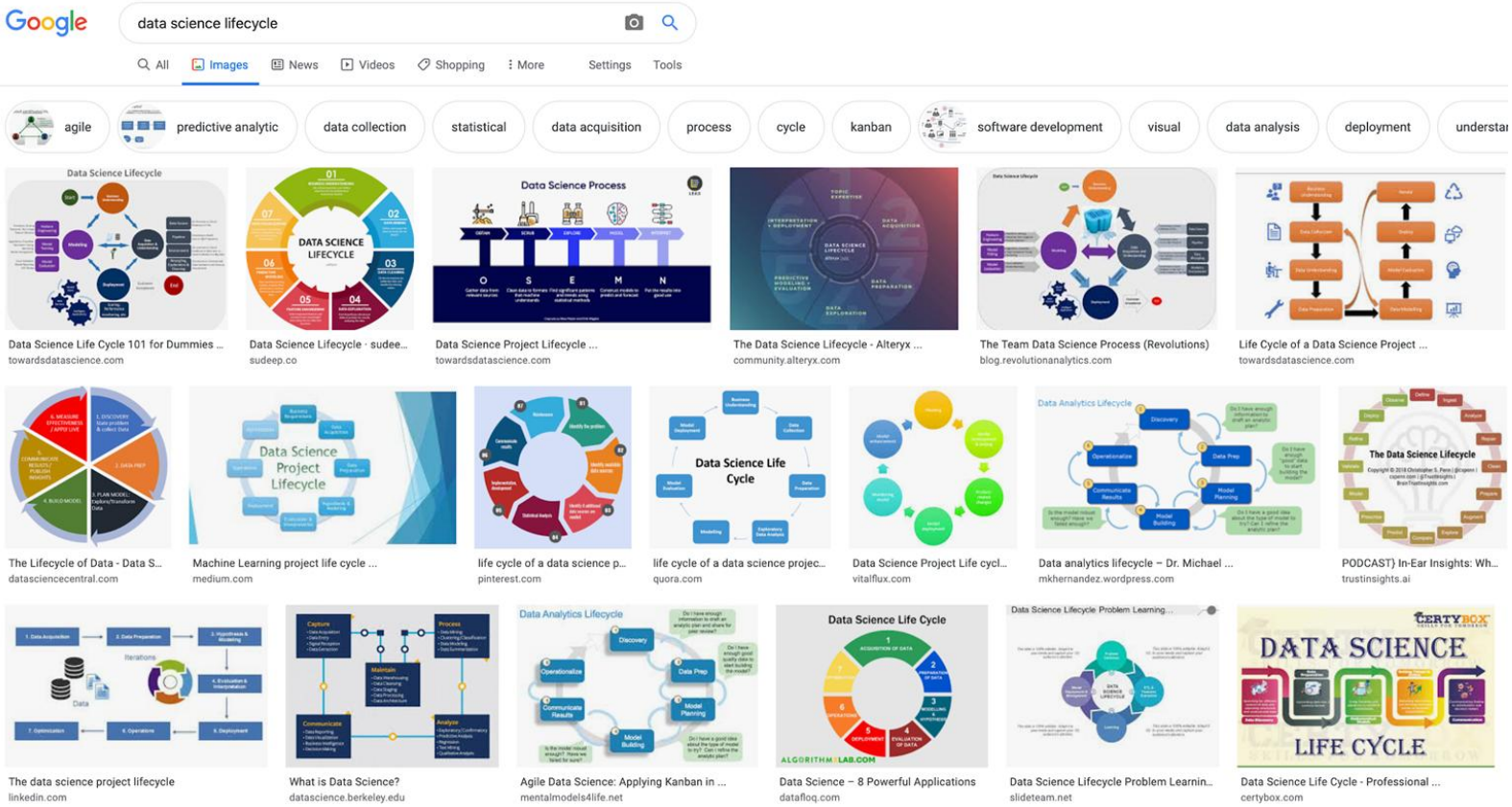
Activity: Buying a new phone

- In pairs, walk through the 6 steps for buying a phone with a budget of 10 million VND:
  - What exactly is your question?
  - What data would you collect, and from where?
  - What problems might you find when exploring it?
  - How would you clean it?
  - How would you analyse it (criteria? weights?)
  - How would you justify your choice to someone else?

What data science is

Data science lifecycle

- The "data science lifecycle" you will see in the wild may be slightly different than the one we teach you, but the core ideas are all the same.

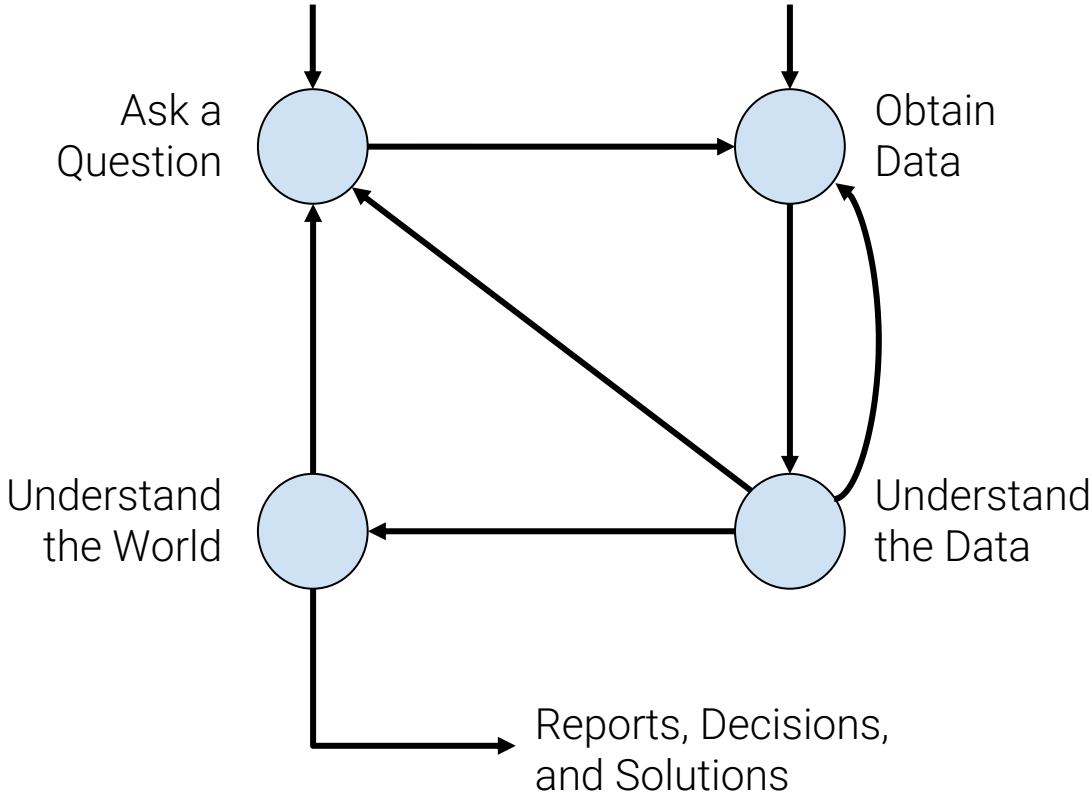


Data science lifecycle

- **Ask** a meaningful question
- **Collect** data
- **Explore** data: understand it and identify problems
- **Preprocess** data
- **Analyse** data
  - *Simple analysis:* computation & visualisation → questions about the **observed** data
  - *Complex analysis:* modelling / machine learning → questions **beyond** the observed data
- **Communicate** results / make decisions

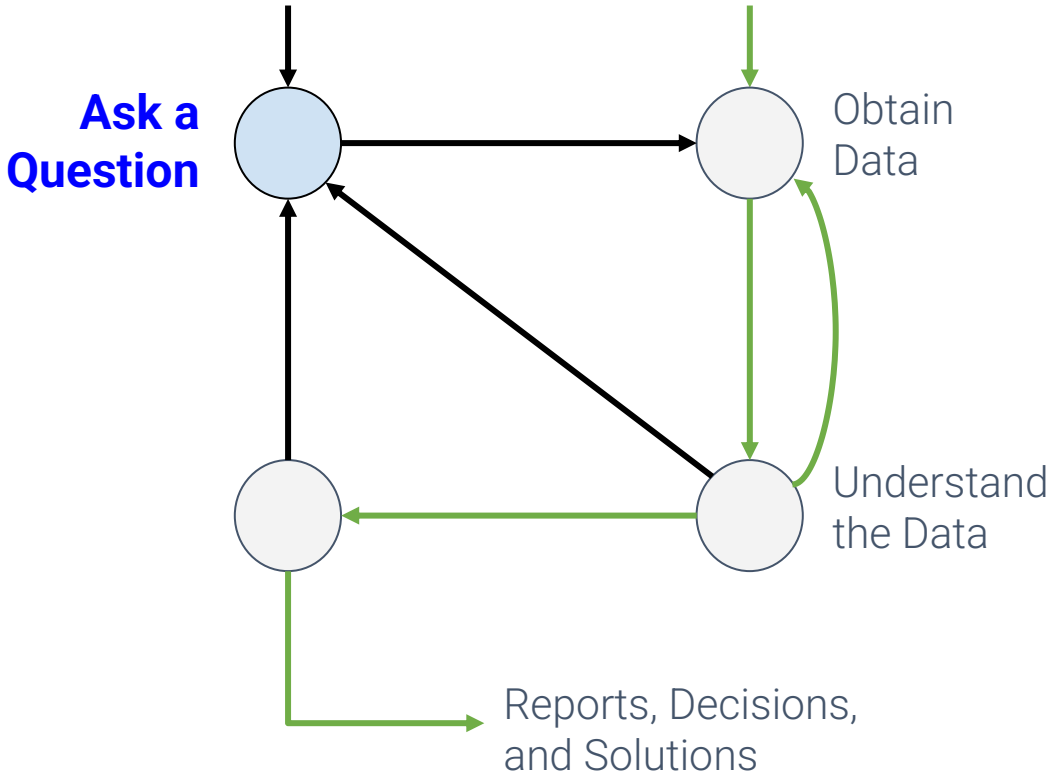
Data science lifecycle

- The data science lifecycle is a high-level description of the data science workflow.
- Note the two distinct entry points!



1. Question/Problem Formulation

- What **problems** should we solve?
- What **hypotheses** should we test?
- What are our **metrics** for success?



1. Question/Problem Formulation

What Makes a Question "Meaningful"?

- Specific and Clear
  - Not "How to improve sales?" but "Which product features drive customer purchases?“
- Answerable with Data
  - Can you collect data to answer it?
- Actionable
  - Will the answer lead to a decision or action?
- Valuable
  - Does solving this matter to stakeholders?

1. Question/Problem Formulation

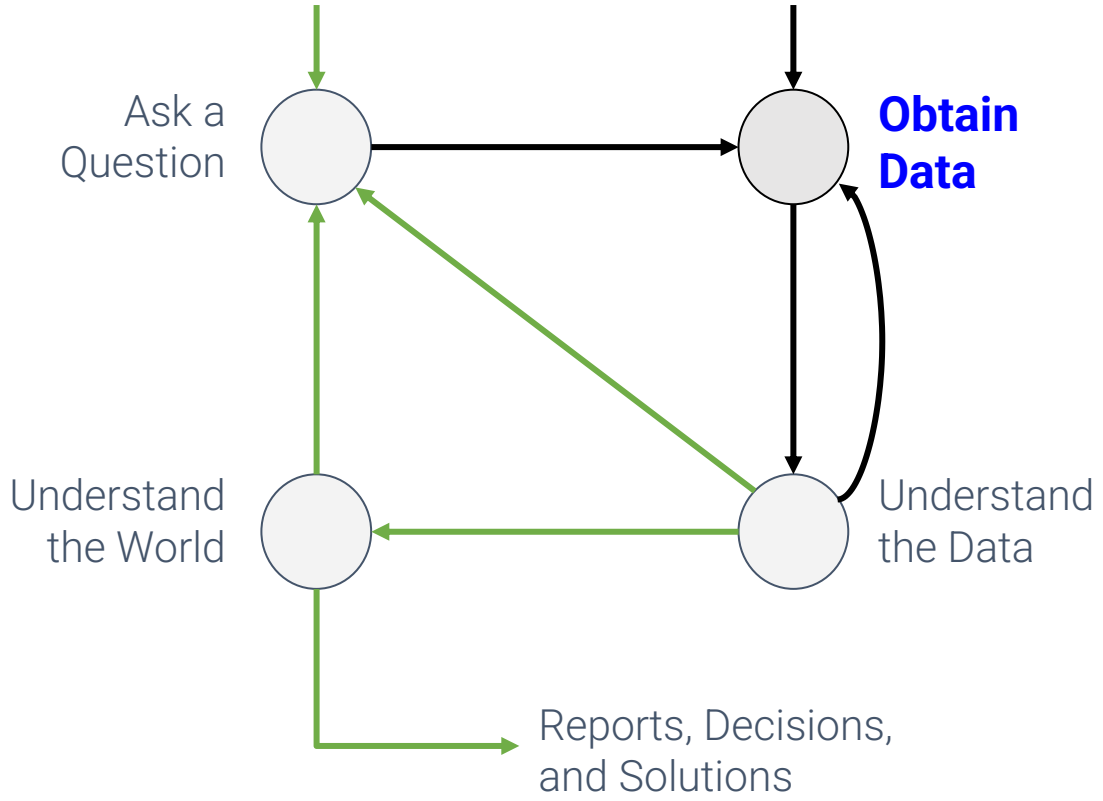
- Poor: "Make the business better with data“
  - Too vague, not actionable
- Better: "Which marketing channel (email, social media, or ads) generates the highest ROI for customer acquisition?
  - "Specific, measurable, actionable
- Poor: "Why are students unhappy?“
  - Too broad, hard to measure
- Better: "What factors (class size, teaching style, workload) correlate most strongly with student satisfaction scores?“
  - Specific variables, measurable outcome

Activity: Fix these questions

- Individually, rewrite each into a meaningful question, then compare with a neighbour:
  1. "Is Grab good?"
  2. "What do students think about the canteen?"
  3. "Is traffic in Ho Chi Minh City bad?"
- For each, name: **the data you'd need** and **the decision it would support**.

2. Data Acquisition and Cleaning

- What data do we **have** and what data do we **need**?
- How can we **sample** more data?
- Is our data **representative** of the population we want to study?



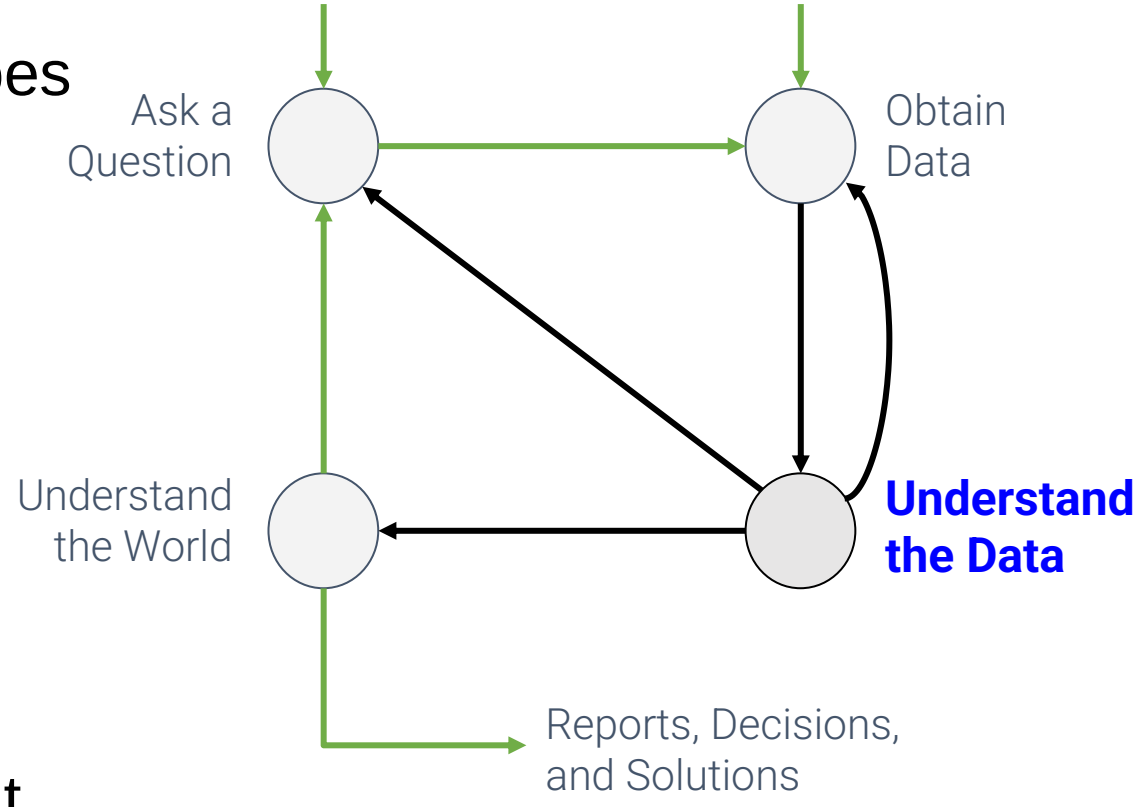
A careful analysis of any problem requires the use of data

2. Data Acquisition and Cleaning

- Data source:
  - **Internal**: company databases, logs
  - **External**: open data, APIs, web
- Collection Type:
  - **Primary***:* you collect it yourself for your purpose (surveys, experiments)
  - **Secondary***:* already collected by others (Kaggle, government portals)
- Quality Factors
  - Accuracy, completeness, consistency, timeliness, relevance
- Ethical Considerations
  - Consent, privacy / personal data, licensing & terms of use

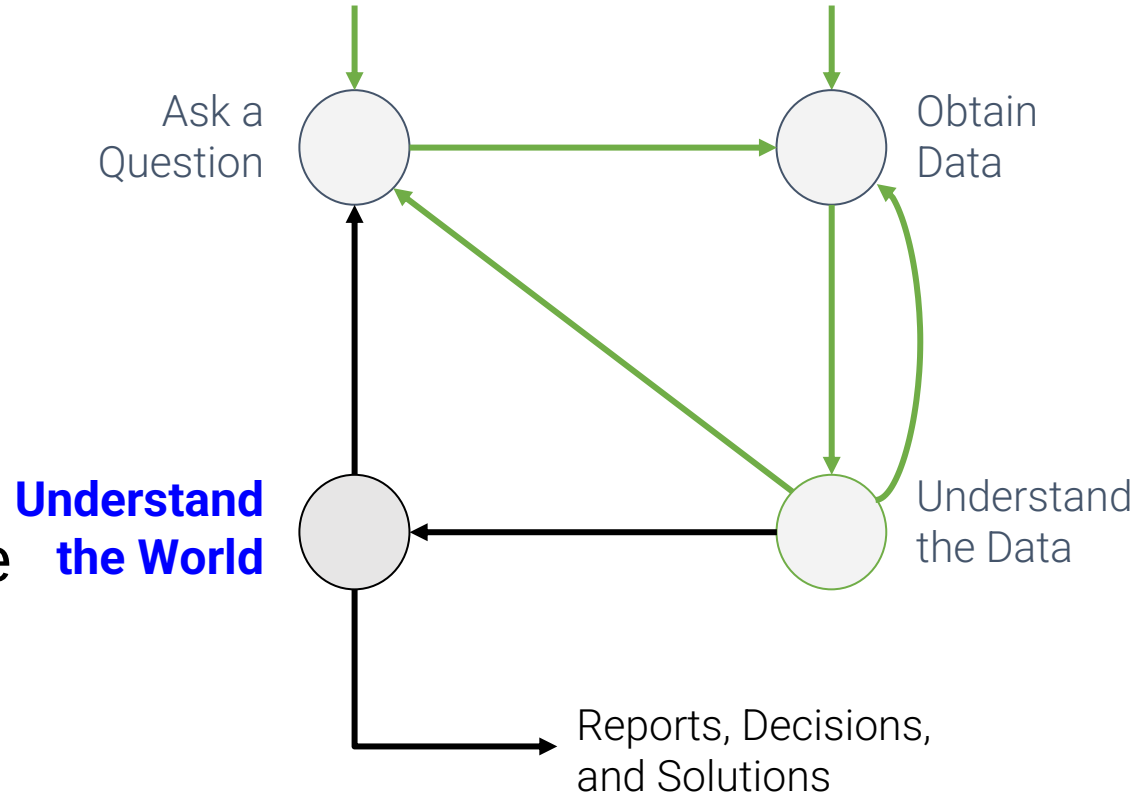
3. Exploratory Data Analysis & Visualization

- How is our data **organized** and what does it contain?
- Do we have **relevant** data?
- What are the biases, anomalies, or other **issues** with the data?
- How do we **transform** it for effective analysis?



4. Prediction and Inference

- What does the data say about the world?
- Does it **answer** our questions or accurately **solve** the problem?
- How **robust** are our conclusions and can we trust the predictions?



Data science lifecycle

- **The process is not linear**
  - It can start from a **question**…
  - …or from **data**: explore first, then identify questions
  - You'll often go **back** to an earlier step, many times
- To do this well, a data scientist needs to:
  - Stay **calm, objective, and honest**, especially when data contradicts what you hoped
  - **Master the software tools** ← *that's this course*

Data science: definitions

- **Data Science** is an interdisciplinary field that leverages *scientific methods*, *algorithms*, and *advanced systems* to extract meaningful knowledge and actionable insights from both structured and unstructured data.

> ***It's the art of turning raw data into strategic, data-driven decisions.***



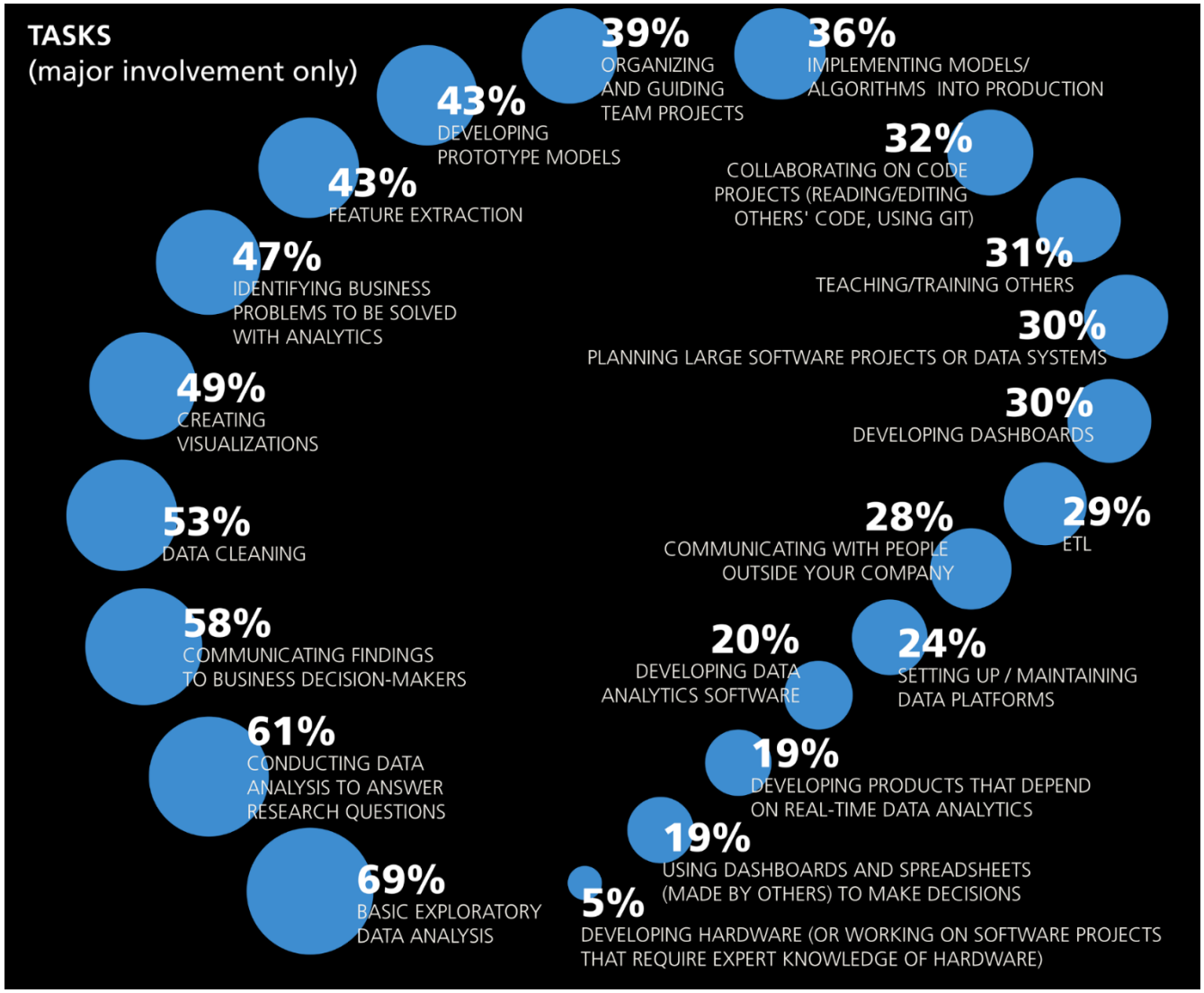
Data science: definitions

- Data Science is the application of data-centric, computational, and inferential thinking to:
  - Understand the world (science)
  - Solve problems (engineering)
- A true mastery of data science requires a deep theoretical understanding and strong grasp of domain expertise

The Interdisciplinary Core

- Data Science sits at the intersection of three major areas, requiring a blend of skills to be effective.
- **Mathematics & Statistics:** foundation for modeling and inference
  - Probability theory
  - Linear algebra
- **Computer Science:** tools and techniques for handling data
  - Programming and algorithms
  - Database management
  - Cloud computing
- **Domain Knowledge:**
  - Understanding the business context
  - Translating problems into data questions

What Do Industry Data Scientists Do?



- The tasks that data scientists say they work on regularly.
- Hmm, can't a large language model (LLM) do a lot of these tasks? Will LLMs replace data scientists?

Source: Based on O'Reilly Data Science Salary Survey

Are LLMs going to replace data scientists?

- Short answer: No one knows for sure, but ...

Step 1	Step 2	Step 3
Gather context and ask insightful questions.	Plan analyses that address your question(s).	Run your chosen analyses.
LLMs cannot deeply understand the **nuanced** desires, abilities, and personalities of complex organizations. At least for now 🙂	LLMs can provide advice as you weigh options. But, the choice of "best" analysis is often **subjective**. Experience matters.	With proper context and instructions, LLMs can help a lot here! But, you must be able to **verify** outputs. Don't do analysis by "vibes".

Are LLMs going to replace data scientists?

- Short answer: No one knows for sure, but ...

Step 1	Step 2	Step 3
Gather context and ask insightful questions.	Plan analyses that address your question(s).	Run your chosen analyses.

- *Many Data homework problems can be answered by LLMs*.
- If LLMs write your answers, you are fully outsourcing Steps 1 and 2, and you never learn to verify the outputs of Step 3.
- In other words, you are setting yourself up for future replacement by LLMs, regardless of your degree/honors/grades.

LLM is this course

- We encourage you to use LLMs appropriately in this course!
- Use of AI-assisted methods and large language models (LLMs), such as ChatGPT, to generate written or code solutions to assignments is prohibited.
  - Copying or closely paraphrasing any text directly a homework or lab into LLM is considered cheating
  - You should break problems into steps and ask for help with each step in your own words

Using LLM in appropriate ways

- “What does this error message mean? <error message> What could I do to fix it?”
- “Pandas is confusing me; what does the following pandas expression do? <weird expression>”
- “How would I convert columns into separate rows in my dataframe?”
- “Can you explain why the regular expression r’[a-z]’ only matches one character? Can you give me a few practice regular expression exercises?”
- If you use an LLM to assist with your exercises, please provide a clear citation. Specify where you used it, what prompt you gave, the response you received, and how you modified or adjusted that response.

Important reminder

- **Remember**: the main goal here is to learn, truly learn
- You can discuss ideas with others as well as consult Internet sources, but your writing and code must be your own,based on your own understanding
- If you violate this rule, you will get 0 score for the course

Data science is hot nowadays

The Era of (Big) Data

~500 million TB/day	**2 ZB → 181 ZB**	**#1 fastest-growing job**
Data created worldwide per day in 2025	Yearly data volume, 2010 → 2025	Big Data Specialists, 2025–2030
IDC Global DataSphere, via Statista	*IDC, via Statista*	*WEF Future of Jobs Report 2025*

***terabyte, petabyte, exabyte, zettabyte**

- Data scientist jobs projected to grow **34%** (2024–2034), vs. ~3% for all jobs.

Data science is hot nowadays

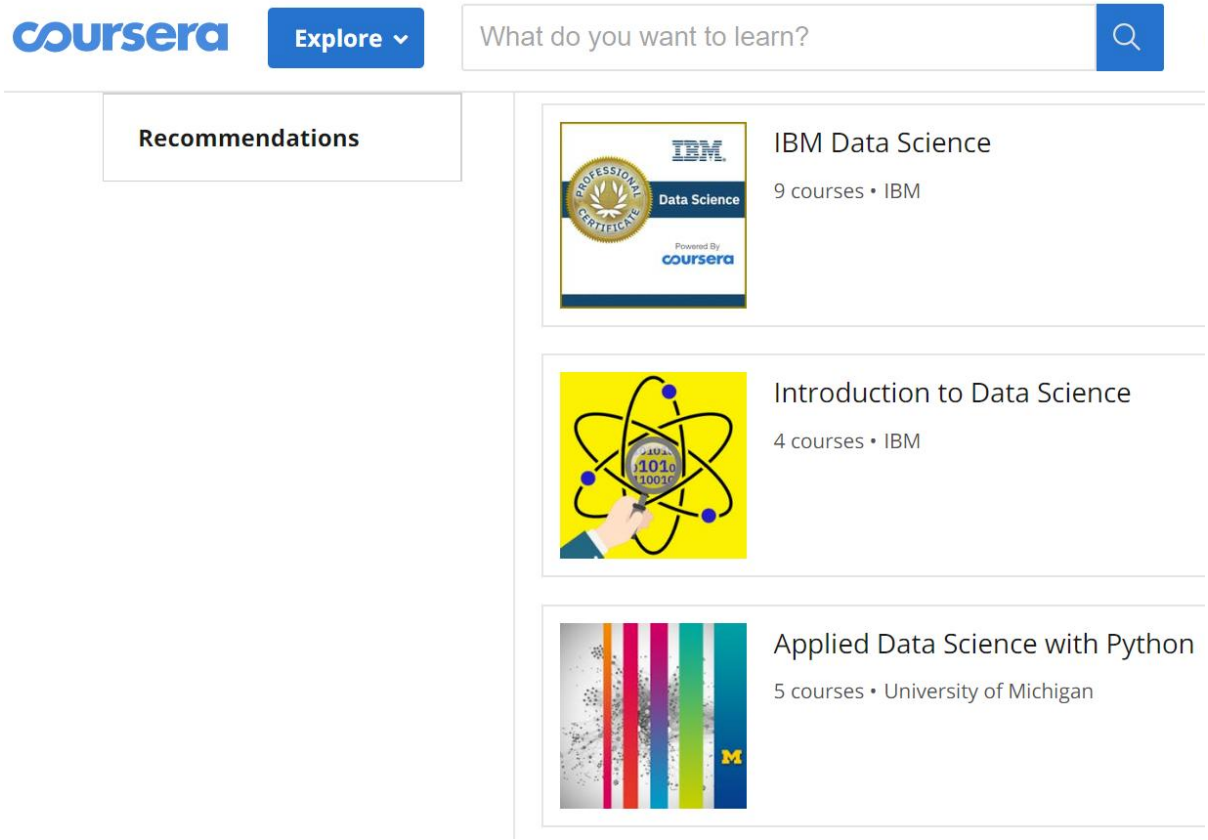
- **Data keeps growing**: from phones, apps, sensors, transactions…
  - Correct and incorrect data are mixed together, and we often can't tell which is which
- **Computers keep getting faster and cheaper**, so big data can be processed in reasonable time
- Organisations that use data well make **better operational and customer decisions**
- In Vietnam: Tiki, Lazada, Shopee, Zalo (VNG), Grab, FPT, MoMo, banks, telcos… all hire data people.

Data science is hot nowadays

- Let's make a tour of its applications in different fields

Recommender system

- Question: what products will the user X like?
- Data: history data of the user X (have liked/bought what products) and other users



Web design for online shop

- Question: how should the website be designed to attract customers?
- Data: interaction data of customers with different design versions of the website



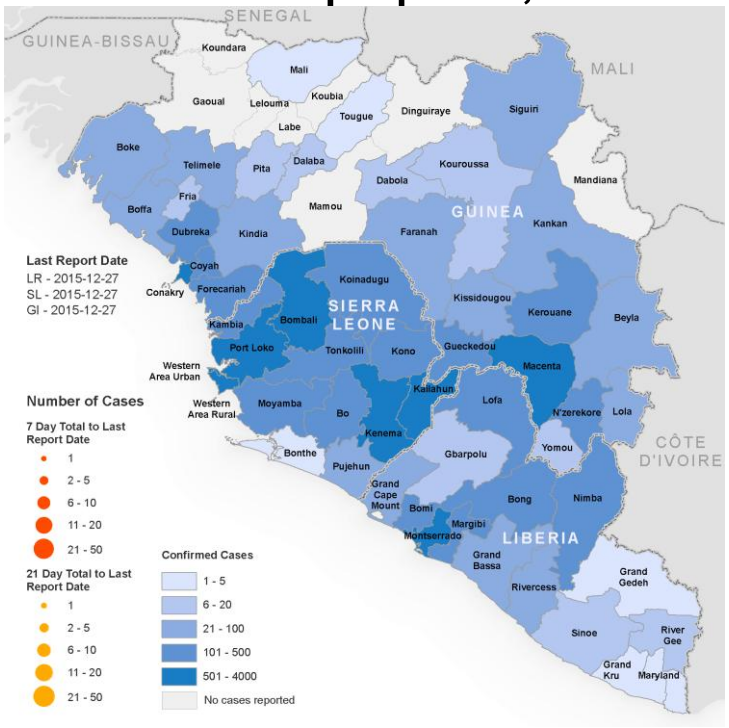
Image source: https://www.optimizely.com/optimizationglossary/ab-testing/

Sentiment analysis

- Question: Does user community respond positively or negatively to the new product X?
- Data: users' reviews in social networks

Epidemic control

- Question: what is the current state of the epidemic X in different locations?
- Data: data about the epidemic X in different locations (which can be collected via newspapers, social networks, …)



http://apps.who.int/ebola/sites/default/files/thumbnails/image/sitrep_casecount_31.png

Precision agriculture

- Question: where in the farm field do we need supply water/fertilizer/…?
- Data: data about the field collected via drone, …



Image source: https://www.dronitech.com/drones-arerevolutionizing-the-future-of-agriculture-farming/

Education improvement

- Question: what aspects of the course X do we need to improve?
- Data: feedback data of students about the course

Criminal investigation

- Question: what crimes were committed by the same offender?
- Data: data about different crimes

Course contents

- In this course, we will learn about tools for doing data science process:
  - Linux commands
  - Git & Github
  - Conda
  - Jupyter notebook
  - Markdown
  - Python
  - Matplotlib
  - Numpy
  - Pandas

Course contents

- You already know some of these tools. We won't re-teach the basics; we'll **level them up** for real data work.

Tool	You already…	In this course we focus on…
Python	Know the basics	Clean, reusable code for data work: functions, modules, comprehensions, error handling, reading CSV/JSON
Jupyter Notebook	Have used it	**Reproducible** notebooks: *Restart & Run All*, clear structure, when to move code into .py files
Markdown	Have used it	**Communicating** results: narrative in notebooks, a good project README
Conda	Have installed it	**Reproducible environments**: environment.yml, pinning versions, sharing with a team

Course contents

Deep dive	Used for
Linux command line	Files, pipes, grep/find, scripts, remote servers
Git & GitHub	Version control and **team** workflow: branches, pull requests, merge conflicts
NumPy	Fast numerical arrays: vectorisation, broadcasting
Pandas	Tabular data: load, clean, reshape, group, merge
Matplotlib	Visualisation for exploring and explaining data

What we learn: tool + principle

- **Principle** (why it works) → **Tool** (how the tool embodies it) → **Practice** (apply both in the lab)

Tool	Principle behind it
Linux CLI	Unix philosophy, automation
Git & GitHub	Snapshots, branching & merging
Conda	Reproducibility: solve "it works on my machine"
Jupyter & md	Code + story, no hidden state
Python	Abstraction & DRY, readability, testing with assert
NumPy	Vectorisation, don’t loop
Pandas	Tidy data; split-apply-combine
Matplotlib	Chart choice, perception, honest plots

Course contents

- After successfully completing this course, you will be able to:
  - Use learned tools
  - to do learned data science process with new data
  - reproducibly
  - and as a team

Beyond the tools: four core principles

Principle	The question to ask yourself
Reproducibility	Can someone else (or me in 6 months) rerun this and get the **same** result?
Verification	How do I know this result is **right**? Do row counts and totals make sense?
Efficiency awareness	Roughly how much work is this code doing? Why is it **slow**?
Communication	Could a **non-expert** understand what I found and why it matters?

Data science lifecycle

- **Ask** a meaningful question
- **Collect** data
- **Explore** data: understand it and identify problems
- **Preprocess** data
- **Analyse** data
  - *Simple analysis:* computation & visualisation → questions about the **observed** data
  - *Complex analysis:* modelling / machine learning → questions **beyond** the observed data
- **Communicate** results / make decisions

Course contents

- **Question**: What is the relationship between this course and the“Introduction to Data Science” course?
- **Answer** (my view):

	Intro to Data Science	**Programming for Data Science**
Focus	The data science **process**	The **tools**, and the principles behind them
Breadth	Covers all steps of the lifecycle	Doesn't try to cover every step
Depth on tools	Basic level	Deep dive into the important tools

Course assessment

- **Homeworks** throughout the course: 20% of the grade
- **Lab exercises** throughout the course: 40% of the grade
- **Group final project**: 40% of the grade
  - 2-3 students / group
  - Use Git & Github to control versions and collaborate
  - Find a public dataset about things your group is interested in, use tools to: explore data, identify meaningful questions which can be answered with this data, preprocess and analyze data to answer each question
  - Present results to Teacher

References

- The “Stopping to Sharpen Your Tools” talk of Brandon Rhodes
- The “The Missing Semester” course of MIT
- The “Computational and Inferential Thinking” book of Berkeley
- Tool documents:
  - Git
  - Conda
  - Jupyter notebook
  - Python
  - Numpy
  - Matplotlib
  - Pandas



THE END